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Optical module having conductive fiber sheet

Abstract

An optical module includes a shell, an elastic sleeve and a conductive fiber sheet. The shell has a groove disposed on an outer wall thereof. The elastic sleeve includes a metal frame and a plurality of metal elastic pieces. The metal frame is sleeved in the groove, and the metal frame is fixedly connected to the shell. The plurality of metal elastic pieces are disposed on an edge of the metal frame and extend to an outside of the metal frame. The conductive fiber sheet is disposed between the shell and the elastic sleeve for connecting the shell and the elastic sleeve.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation Application of PCT/CN2021/100971 filed on Jun. 18, 2021, which claims priority to Chinese Patent Application No. 202010811010.9 filed on Aug. 13, 2020, and Chinese Patent Application No. 202021942876.5 filed on Sep. 8, 2020, which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates to the field of optical communication technologies, and in particular, to an optical module.

BACKGROUND

(2) Optical communication technologies are used in new services and application scenarios such as cloud computing, mobile internet, and video conferencing. In optical communication, an optical module is a tool for achieving interconversion between an optical signal and an electrical signal, and is one of key devices in an optical communication device. The optical module generally includes a light emitting device and a light receiving device. The light emitting device is configured

to convert an electrical signal into an optical signal, and transmit the optical signal out through an optical fiber, and the light receiving device is configured to convert an optical signal transmitted from the optical fiber into an electrical signal.

SUMMARY

(3) In an aspect, an optical module is provided. The optical module includes a shell, an elastic sleeve and a conductive fiber sheet. The shell has a groove disposed on an outer wall thereof. The elastic sleeve includes a metal frame and a plurality of metal elastic pieces. The metal frame is sleeved in the groove, and the metal frame is fixedly connected to the shell. The plurality of metal elastic pieces are disposed on an edge of the metal frame and extend to an outside of the metal frame. The conductive fiber sheet is disposed between the shell and the elastic sleeve for connecting the shell and the elastic sleeve.

(4) In another aspect, an optical module is provided. The optical module includes a shell, an unlocking component and at least one conductive fiber sheet. The shell has at least one recessed area disposed on an outer wall thereof. The unlocking component is connected to the at least one recessed area. The at least one conductive fiber sheet is disposed in a corresponding recessed area and located between the unlocking component and the shell, and the at least one conductive fiber sheet is connected to the shell and the unlocking component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe technical solutions in the present disclosure more clearly, accompanying drawings to be used in some embodiments of the present disclosure will be introduced briefly below. Obviously, the accompanying drawings to be described below are merely accompanying drawings of some embodiments of the present disclosure, and a person of ordinary skill in the art may obtain other drawings according to these drawings. In addition, the accompanying drawings in the following description may be regarded as schematic diagrams, and are not limitations on actual sizes of products, actual processes of methods and actual timings of signals involved in the embodiments of the present disclosure.

(2) FIG. 1 is a diagram showing a connection relationship of an optical communication system, in accordance with some embodiments;

(3) FIG. 2 is a structural diagram of an optical network terminal, in accordance with some embodiments;

(4) FIG. 3 is an exploded structural diagram of an optical module, in accordance with some embodiments;

(5) FIG. 4 is an assembly diagram of a shell and an elastic sleeve in an optical module, in accordance with some embodiments;

(6) FIG. 5 is an exploded view of a shell, an elastic sleeve, and a conductive fiber sheet in an optical module, in accordance with some embodiments;

(7) FIG. 6 is a structural diagram of an elastic sleeve in an optical module, in accordance with some embodiments;

(8) FIG. 7 is a diagram of a shell in an optical module from another perspective, in accordance with some embodiments;

(9) FIG. 8 is a structural diagram of a conductive fiber sheet in an optical module, in accordance with some embodiments;

(10) FIG. 9 is an assembly diagram of a conductive fiber sheet and a shell in an optical module, in accordance with some embodiments;

(11) FIG. 10 is an assembly diagram of an elastic sleeve, a conductive fiber sheet and a shell in an optical module, in accordance with some embodiments;

- (12) FIG. 11 is a side view of an assembly of an elastic sleeve, a conductive fiber sheet and a shell in an optical module, in accordance with some embodiments;
- (13) FIG. 12 is an enlarged view of a portion A in FIG. 11;
- (14) FIG. 13 is an assembly diagram of a shell and an unlocking component in an optical module, in accordance with some embodiments;
- (15) FIG. 14 is an exploded structural diagram of a shell and an unlocking component in an optical module, in accordance with some embodiments;
- (16) FIG. 15 is an assembly diagram of a shell and an unlocking component in an optical module, in accordance with some embodiments;
- (17) FIG. 16 is an assembly diagram of a conductive pad and a shell in an optical module, in accordance with some embodiments;
- (18) FIG. 17 is an exploded diagram of a shell, a conductive pad, and an unlocking component in an optical module, in accordance with some embodiments;
- (19) FIG. 18 is a sectional view of a conductive pad, an unlocking component, and a shell in an optical module, in accordance with some embodiments; and
- (20) FIG. 19 is another structural diagram of a shell in an optical module, in accordance with some embodiments.

DETAILED DESCRIPTION

- (21) Technical solutions in some embodiments of the present disclosure will be described clearly and completely below with reference to the accompanying drawings. Obviously, the described embodiments are merely some but not all embodiments of the present disclosure. All other embodiments obtained on a basis of the embodiments of the present disclosure by a person of ordinary skill in the art shall be included in the protection scope of the present disclosure.
- (22) Unless the context requires otherwise, throughout the specification and the claims, the term “comprise” and other forms thereof such as the third-person singular form “comprises” and the present participle form “comprising” are construed as being open and inclusive, meaning “including, but not limited to”. In the description, the terms such as “one embodiment”, “some embodiments”, “exemplary embodiments”, “example”, “specific example” or “some examples” are intended to indicate that specific features, structures, materials, or characteristics related to the embodiment(s) or example(s) are included in at least one embodiment or example of the present disclosure. Schematic representations of the above terms do not necessarily refer to the same embodiment(s) or example(s). In addition, the specific features, structures, materials, or characteristics described herein may be included in any one or more embodiments or examples in any suitable manner.
- (23) Hereinafter, the terms “first” and “second” are only used for descriptive purposes, and cannot be construed as indicating or implying the relative importance or implicitly indicating the number of indicated technical features. Thus, features defined with “first” and “second” may explicitly or implicitly include one or more of the features. In the description of the embodiments of the present disclosure, the term “a plurality of” means two or more unless otherwise specified.
- (24) In the description of some embodiments, the terms “coupled” and “connected” and their extensions may be used. For example, the term “connected” may be used in the description of some embodiments to indicate that two or more components are in direct physical or electrical contact with each other. For another example, the term “coupled” may be used in the description of some embodiments to indicate that two or more components are in direct physical or electrical contact. However, the term “coupled” or “communicatively coupled” may also mean that two or more components are not in direct contact with each other, but still cooperate or interact with each other. The embodiments disclosed herein are not necessarily limited to the contents herein.
- (25) The phrase “at least one of A, B and C” has a same meaning as the phrase “at least one of A, B or C”, and both include the following combinations of A, B and C: only A, only B, only C, a combination of A and B, a combination of A and C, a combination of B and C, and a combination

of A, B and C.

(26) The phrase “A and/or B” includes the following three combinations: only A, only B, and a combination of A and B.

(27) The use of the phrase “applicable to” or “configured to” herein means an open and inclusive language, which does not exclude devices that are applicable to or configured to perform additional tasks or steps.

(28) The term such as “about”, “substantially” or “approximately” as used herein includes a stated value and an average value within an acceptable range of deviation of a particular value. The acceptable range of deviation is determined by a person of ordinary skill in the art, considering measurement in question and errors associated with measurement of a particular quantity (i.e., limitations of a measurement system).

(29) In an optical communication system, an optical signal is used to carry information to be transmitted, and the optical signal carrying the information is transmitted to an information processing device such as a computer through an information transmission device such as an optical fiber or an optical waveguide, so as to achieve transmission of the information. Since light has a characteristic of passive transmission when being transmitted through the optical fiber or the optical waveguide, low-cost and low-loss information transmission may be achieved. In addition, since a signal transmitted by the information transmission device such as the optical fiber or the optical waveguide is an optical signal, and a signal that can be recognized and processed by the information processing device such as a computer is an electrical signal, in order to establish information connection between the information transmission device such as the optical fiber or the optical waveguide and the information processing device such as the computer, there is a need to achieve interconversion between the electrical signal and the optical signal.

(30) In the field of optical communication technology, an optical module may be used to achieve the interconversion between the optical signal and the electrical signal. The optical module includes an optical port and an electrical port. The optical module achieves optical communication with the information transmission device such as the optical fiber or the optical waveguide through the optical port, and achieves electrical connection with an optical network terminal (e.g., an optical modem) through the electrical port. The electrical connection is mainly used to achieve power supply, transmission of an I2C signal, transmission of data information and grounding. The optical network terminal transmits the electrical signal to the information processing device such as the computer through a network cable or wireless fidelity (Wi-Fi).

(31) FIG. 1 is a diagram showing a connection relationship of an optical communication system. As shown in FIG. 1, the optical communication system includes a remote server **1000**, a local information processing device **2000**, an optical network terminal **100**, an optical module **200**, an optical fiber **101** and a network cable **103**.

(32) One end of the optical fiber **101** is connected to the remote server **1000**, and the other end of the optical fiber **101** is connected to the optical network terminal **100** through the optical module **200**. The optical fiber itself may support long-distance signal transmission, such as several-kilometer (six-kilometer to eight-kilometer) signal transmission. On this basis, infinite-distance transmission may be realized theoretically if a repeater is used. Therefore, in a typical optical communication system, a distance between the remote server **1000** and the optical network terminal **100** may typically reach several kilometers, tens of kilometers, or hundreds of kilometers.

(33) One end of the network cable **103** is connected to the local information processing device **2000**, and the other end of the network cable **103** is connected to the optical network terminal **100**. The local information processing device **2000** may be any one or more of the following devices: a router, a switch, a computer, a mobile phone, a tablet computer, a television, etc.

(34) A physical distance between the remote server **1000** and the optical network terminal **100** is greater than a physical distance between the local information processing device **2000** and the optical network terminal **100**. Connection between the local information processing device **2000**

and the remote server **1000** is implemented by the optical fiber **101** and the network cable **103**, and connection between the optical fiber **101** and the network cable **103** is implemented by the optical module **200** and the optical network terminal **100**.

(35) The optical module **200** includes an optical port and an electrical port. The optical port is configured to be connected to the optical fiber **101**, so that bidirectional optical signal connection is established between the optical module **200** and the optical fiber **101**. The electrical port is configured to be connected to the optical network terminal **100**, so that bidirectional electrical signal connection is established between the optical module **200** and the optical network terminal **100**. The optical module **200** achieves the interconversion between the optical signal and the electrical signal, so that information connection between the optical fiber **101** and the optical network terminal **100** is established. For example, an optical signal from the optical fiber **101** is converted into an electrical signal by the optical module **200**, and then the electrical signal is input into the optical network terminal **100**; an electrical signal from the optical network terminal **100** is converted into an optical signal by the optical module **200**, and then the optical signal is input into the optical fiber **101**. Since the optical module **200** is a tool for achieving the interconversion between the optical signal and the electrical signal, and does not have a function of processing data, the information does not change in the above photoelectric conversion process.

(36) The optical network terminal **100** includes a substantially cuboid housing, and an optical module interface **102** and a network cable interface **104** that are disposed on the housing. The optical module interface **102** is configured to be connected to the optical module **200**, so that the bidirectional electrical signal connection between the optical network terminal **100** and the optical module **200** is established. The network cable interface **104** is configured to be connected to the network cable **103**, so that bidirectional electrical signal connection between the optical network terminal **100** and the network cable **103** is established. Connection between the optical module **200** and the network cable **103** is established through the optical network terminal **100**. For example, the optical network terminal transmits an electrical signal from the optical module **200** to the network cable, and transmits an electrical signal from the network cable **103** to the optical module **200**. Therefore, the optical network terminal **100**, as a master monitor of the optical module **200**, may monitor operation of the optical module **200**. In addition to the optical network terminal **100**, the master monitor of the optical module **200** may further include an optical line terminal (OLT).

(37) The remote server **1000** establishes a bidirectional signal transmission channel with the local information processing device **2000** through the optical fiber **101**, the optical module **200**, the optical network terminal **100** and the network cable **103**.

(38) FIG. 2 is a structural diagram of an optical network terminal. In order to clearly show a connection relationship between the optical module **200** and the optical network terminal **100**, FIG. 2 only shows a structure of the optical network terminal **100** related to the optical module **200**. As shown in FIG. 2, the optical network terminal **100** further includes a circuit board **105** disposed in the housing, a cage **106** disposed on a surface of the circuit board **105**, an electrical connector disposed inside the cage **106**, and a heat sink **107** disposed on the cage **106**. The electrical connector is configured to be connected to the electrical port of the optical module **200**. The heat sink **107** has protruding portions such as fins for increasing a heat dissipation area.

(39) The optical module **200** is inserted into the cage **106** of the optical network terminal **100**, the optical module **200** is fixed by the cage **106**, and heat generated by the optical module **200** is conducted to the cage **106** and is diffused through the heat sink **107**. After the optical module **200** is inserted into the cage **106**, the electrical port of the optical module **200** is connected to the electrical connector inside the cage **106**, so that the bidirectional electrical signal connection between the optical module **200** and the optical network terminal **100** is established. In addition, the optical port of the optical module **200** is connected to the optical fiber **101**, so that the bidirectional optical signal connection between the optical module **200** and the optical fiber **101** is established.

(40) FIG. 3 is an exploded structural diagram of an optical module according to some

embodiments. As shown in FIG. 3, the optical module **200** includes a shell **210**, a circuit board **600**, a light emitting device **700** and a light receiving device **800** that are disposed in the shell **210**.

(41) The shell **210** includes an upper shell **201** and a lower shell **202**. The upper shell **201** covers the lower shell **202** to form the shell **210** with two openings, and an outer contour of the shell **210** is generally in a cuboid shape.

(42) In some embodiments, the lower shell **202** includes a bottom plate **2021** and two lower side plates **2022** located on two sides of the bottom plate **2021** and disposed perpendicular to the bottom plate **2021**. The upper shell **201** includes a cover plate **2011**, and the cover plate **2011** covers the two lower side plates **2022** of the lower shell **202** to form the shell **210**.

(43) In some embodiments, as shown in FIG. 3, the lower shell **202** includes the bottom plate **2021** and the two lower side plates **2022** located on the two sides of the bottom plate **2021** and disposed perpendicular to the bottom plate **2021**. The upper shell **201** includes the cover plate **2011** and two upper side plates **2012** located on two sides of the cover plate **2011** and disposed perpendicular to the cover plate **2011**. The two upper side plates **2012** are combined with the two lower side plates **2022**, so that the upper shell **201** covers the lower shell **202**.

(44) A direction in which a connecting line between the two openings **204** and **205** is located may be the same as a longitudinal direction of the optical module **200**, or may not be the same as the longitudinal direction of the optical module **200**. For example, the opening **204** is located at an end (a right end in FIG. 4) of the optical module **200**, and the opening **205** is also located at an end (a left end in FIG. 4) of the optical module **200**. Alternatively, the opening **204** is located at an end of the optical module **200**, and the opening **205** is located at a side of the optical module **200**. The opening **204** is the electrical port. A connecting finger **601** of the circuit board **600** extends from the electrical port **204**, and is inserted into the master monitor (e.g., the optical network terminal **100**). The opening **205** is the optical port, and is configured to be connected to an external optical fiber **101**, so that the external optical fiber **101** is connected to the light emitting device **700** and the light receiving device **800** inside the optical module **200**.

(45) By using an assembly manner of combining the upper shell **201** and the lower shell **202**, it is convenient to install devices such as the circuit board **600**, the light emitting device **700** and the light receiving device **800** into the shell **210**, and the upper shell **201** and the lower shell **202** may form encapsulation protection for these devices. In addition, when the devices such as the circuit board **600**, the light emitting device **700** and the light receiving device **800** are assembled, it is convenient to arrange positioning components, heat dissipation components and electromagnetic shielding components of these devices, which is conducive to implementation of automated production.

(46) In some embodiments, the upper shell **201** and the lower shell **202** are generally made of a metal material, which facilitates electromagnetic shielding and heat dissipation.

(47) The circuit board **600** includes a circuit wire, electronic elements and chips. Through the circuit wire, the electronic elements and the chips are connected together according to a circuit design, so as to implement functions such as power supply, transmission of electrical signals and grounding. The electronic elements include, for example, a capacitor, a resistor, a triode, and a metal-oxide-semiconductor field-effect transistor (MOSFET). The chips include, for example, a microcontroller unit (MCU), a laser driver chip, a limiting amplifier, a clock and data recovery (CDR) chip, a power management chip and a digital signal processing (DSP) chip.

(48) The circuit board **600** further includes the connecting finger **601** formed on an end surface thereof, and the connecting finger **601** is composed of a plurality of independent pins. The circuit board **600** is inserted into the cage **106**, and is conductively connected to the electrical connector in the cage **106** through the connecting finger **601**. The connecting finger **601** may be disposed on only a surface (e.g., an upper surface shown in FIG. 3) of the circuit board **600**, or may be disposed on both upper and lower surfaces of the circuit board **600** to adapt to an occasion with a demand for a large number of pins. The connecting finger **601** is configured to establish electrical connection

with the master monitor to achieve power supply, grounding, transmission of an I2C signal, transmission of a data signal, etc.

(49) The circuit board **600** is generally a rigid circuit board, and the rigid circuit board may also implement a load-bearing function due to its relatively hard material. For example, the rigid circuit board may stably bear the electronic elements and the chips. When the light emitting device **700** and the light receiving device **800** are located on the circuit board **600**, the rigid circuit board may further provide a stable bearing. The rigid circuit board may further be inserted into the electrical connector in the cage of the master monitor.

(50) Of course, flexible circuit boards are also used in some optical modules. A flexible circuit board is generally used in conjunction with the rigid circuit board as a supplement to the rigid circuit board. For example, the rigid circuit board is connected to the light emitting device **700** and the light receiving device **800** by means of the flexible circuit board.

(51) The light emitting device **700** is configured to convert an electrical signal into an optical signal, and the light receiving device **800** is configured to convert an optical signal into an electrical signal. The light emitting device **700** and the light receiving device **800** may adopt a transistor outline (TO) package (i.e., coaxial package) or a chip on board (COB) package. The light emitting device **700** and the light receiving device **800** in FIG. 3 adopt the COB package.

(52) In some embodiments, as shown in FIG. 3, the optical module **200** further includes a first optical fiber ribbon **901**, a second optical fiber ribbon **902** and an optical fiber adapter **900**. One end of the first optical fiber ribbon **901** is connected to the light emitting device **700**, and the other end thereof is connected to the optical fiber adapter **900**. One end of the second optical fiber ribbon **902** is connected to the light receiving device **800**, and the other end thereof is connected to the optical fiber adapter **900**. The optical fiber adapter **900** is configured to be connected to the external optical fiber **101**. The light emitting device **700** is optically connected to the external optical fiber **101** through the first optical fiber ribbon **901**, and the light receiving device **800** is optically connected to the external optical fiber **101** through the second optical fiber ribbon **902**. An optical signal emitted by the light emitting device **700** is transmitted to the external optical fiber **101** through the first optical fiber ribbon **901**, and an optical signal from the external optical fiber **101** is transmitted to the light receiving device **800** through the second optical fiber ribbon **902**.

(53) In order to prevent electromagnetic waves generated by the optical module from leaking from a gap between the shell of the optical module and the cage of the master monitor (e.g., the cage **106** of the optical network terminal **100**), a conductive member is disposed between the shell of the optical module and the cage of the master monitor, so as to eliminate the gap between the shell of the optical module and the cage of the master monitor and achieve expected electromagnetic shielding effect. FIG. 4 is an assembly diagram of a shell and an elastic sleeve in an optical module according to some embodiments. FIG. 5 is an exploded view of a shell, an elastic sleeve, and a conductive fiber sheet in an optical module according to some embodiments. As shown in FIGS. 4 and 5, the optical module **200** further includes an elastic sleeve **300**. The shell **210** has a groove **203**, and the groove **203** is disposed on an outer wall of the shell **210** and surrounds the outer wall of the shell **210**. The elastic sleeve **300** is sleeved in the groove **203** of the shell **210** and fixed with the bottom plate **2021** of the lower shell **202**. The elastic sleeve **300** is configured to connect the shell **210** and the cage **106**, so as to ensure sufficient contact between the shell **210** of the optical module **200** and the cage **106** of the optical network terminal **100**, and block the gap between the shell **210** of the optical module **200** and the cage **106** of the optical network terminal **100**.

(54) In some embodiments, the groove **203** includes a first groove **2031** and a second groove **2032**. The first groove **2031** is disposed on an outer wall of the upper shell **201**. That is, the first groove **2031** is disposed on outer walls of the cover plate **2011** and the two upper side plates **2012**. The second groove **2032** is disposed on an outer wall of the lower shell **202**. That is, the second groove **2032** is disposed on outer walls of the bottom plate **2021** and the two lower side plates **2022**.

(55) When the optical module **200** is inserted into the cage **106**, the elastic sleeve **300** is in contact

with an inner surface of the cage **106**, so as to block the gap between the shell **210** of the optical module **200** and the inner surface of the cage **106** of the optical network terminal **100**. That is, a sealed cavity is formed between the shell **210** of the optical module **200** and the cage **106** of the optical network terminal **100** through the elastic sleeve **300**, and electromagnetic waves generated by the optical module **200** are shielded in the sealed cavity, thereby preventing the electromagnetic waves from leaking to an outside of the cage **106** of the optical network terminal **100**.

(56) FIG. **6** is a structural diagram of an elastic sleeve **300** in an optical module according to some embodiments. As shown in FIG. **6**, the elastic sleeve **300** includes a metal frame **301** and a plurality of metal elastic pieces **302**. The metal frame **301** is sleeved in the groove **203** of the shell **210** and is in sufficient contact with the shell **210**. The plurality of metal elastic pieces **302** are uniformly disposed on an edge of the metal frame **301**, and outer surfaces thereof are in contact with the inner surface of the cage **106**. When the optical module **200** is inserted into the cage **106** of the optical network terminal **100**, the plurality of metal elastic pieces **302** are squeezed by the inner surface of the cage **106** and thus elastically deformed, and the plurality of metal elastic pieces **302** are stuck at a socket of the cage **106**, so as to block the gap between the shell **210** of the optical module **200** and the cage **106** of the optical network terminal **100**.

(57) The metal frame **301** includes a first side plate **3011**, a second side plate **3012**, a third side plate **3013**, a fourth side plate **3014** and a fifth side plate **3015**. One end of the first side plate **3011** is connected to the second side plate **3012**, and the other end of the first side plate **3011** is connected to the third side plate **3013**. An end of the second side plate **3012** away from the first side plate **3011** is connected to the fourth side plate **3014**, and an end of the third side plate **3013** away from the first side plate **3011** is connected to the fifth side plate **3015**. The second side plate **3012** and the third side plate **3013** are symmetrically arranged with respect to the first side plate **3011**. That is, the second side plate **3012** and the third side plate **3013** are disposed on a same side of the first side plate **3011**, and an included angle $\alpha_{\text{sub.1}}$ between the second side plate **3012** and the first side plate **3011** is equal to an included angle $\alpha_{\text{sub.2}}$ between the third side plate **3013** and the first side plate **3011**. The fourth side plate **3014** and the fifth side plate **3015** are symmetrically arranged with respect to the first side plate **3011**. That is, the fourth side plate **3014** and the fifth side plate **3015** both extend toward an inside of the elastic sleeve **300**, and an included angle $\alpha_{\text{sub.3}}$ between the fourth side plate **3014** and the second side plate **3012** is equal to an included angle $\alpha_{\text{sub.4}}$ between the fifth side plate **3015** and the third side plate **3013**. The fourth side plate **3014** and the fifth side plate **3015** are located in a same plane, and there is a gap between the fourth side plate **3014** and the fifth side plate **3015**. That is to say, the first side plate **3011**, the second side plate **3012**, the third side plate **3013**, the fourth side plate **3014** and the fifth side plate **3015** constitute a C-shaped metal member, and the metal frame **301** may be sleeved in the groove **203** through the gap between the fourth side plate **3014** and the fifth side plate **3015**.

(58) The two upper side plates **2012** are a first upper side plate **2012A** and a second upper side plate **2012B**, and the two lower side plates **2022** are a first lower side plate **2022A** and a second lower side plate **2022B**. When the metal frame **301** is sleeved in the groove **203**, the first side plate **3011** is clamped into a groove on the cover plate **2011** of the upper shell **201**, the second side plate **3012** is clamped into grooves on the first upper side plate **2012A** of the upper shell **201** and the first lower side plate **2022A** of the lower shell **202**, and the third side plate **3013** is clamped into grooves on the second upper side plate **2012B** of the upper shell **201** and the second lower side plate **2022B** of the lower shell **202**. The fourth side plate **3014** is clamped into a groove on the bottom plate **2021** of the lower shell **202**, and the fifth side plate **3015** is clamped into the groove on the bottom plate **2021** of the lower shell **202**.

(59) FIG. **7** is a diagram of a shell in an optical module from another perspective according to some embodiments. As shown in FIG. **7**, the shell **210** has two clamping grooves **206**, which are a first clamping groove **206A** and a second clamping groove **206B**. The first clamping groove **206A** and the second clamping groove **206B** are both disposed on the bottom plate **2021** of the lower shell

202 and located in the second groove **2032**. The first clamping groove **206A** and the second clamping groove **206B** are symmetrically arranged with respect to a symmetry axis L of the bottom plate **2021**.

(60) The metal frame **301** further includes two elastic fasteners **3016**, which are a first elastic fastener **3016A** and a second elastic fastener **3016B**. The first elastic fastener **3016A** is disposed on the fourth side plate **3014**, the second elastic fastener **3016B** is disposed on the fifth side plate **3015**, and an elastic fastener **3016** is fixedly connected to a clamping groove **206**. When the metal frame **301** is clamped into the groove **203** of the shell **210**, the first elastic fastener **3016A** on the fourth side plate **3014** and the second elastic fastener **3016B** on the fifth side plate **3015** are respectively clamped into the first clamping groove **206A** and the second clamping groove **206B** on the bottom plate **2021** of the lower shell **202**, so as to achieve fixing of the metal frame **301** and the bottom plate **2021** of the lower shell **202**. As a result, the upper shell **201** and the lower shell **202** are wrapped in the metal frame **301**.

(61) In some embodiments, elastic fasteners **3016** on the fourth side plate **3014** and the fifth side plate **3015** of the metal frame **301** may be made by stamping process, so that an end of the first elastic fastener **3016A** is connected to the fourth side plate **3014**, and an end of the second elastic fastener **3016B** is connected to the fifth side plate **3015**. After the elastic fastener **3016** is clamped into the clamping groove **206** on the bottom plate **2021** of the lower shell **202**, the elastic fastener **3016** may be pressed to be separated from the clamping groove **206**, so that the elastic sleeve **300** is easily disassembled.

(62) The plurality of metal elastic pieces **302** are uniformly disposed on edges of the first side plate **3011**, the second side plate **3012**, the third side plate **3013**, the fourth side plate **3014** and the fifth side plate **3015** of the metal frame **301**. The plurality of metal elastic pieces **302** extend to an outside of the elastic sleeve **300** along the edges of the first side plate **3011**, the second side plate **3012**, the third side plate **3013**, the fourth side plate **3014** and the fifth side plate **3015**, and are arranged in an umbrella shape.

(63) It will be noted that the number of metal elastic pieces **302** disposed on the edges of the first side plate **3011**, the second side plate **3012**, the third side plate **3013**, the fourth side plate **3014** and the fifth side plate **3015** is not limited. According to the actual situation, as many metal elastic pieces **302** as possible may be provided to increase a contact area between the elastic sleeve **300** and the cage **106**.

(64) After the elastic sleeve **300** is sleeved in the groove **203** of the shell **210**, the metal frame **301** of the elastic sleeve **300** is in contact with the outer walls of the upper shell **201** and the lower shell **202**, and the metal elastic pieces **302** of the elastic sleeve **300** are in contact with the inner surface of the cage **106**, so that sealing between the shell **210** and the cage **106** is achieved. The elastic sleeve **300** is a metal member, and a sealed metal cavity is formed among the shell **210**, the elastic sleeve **300** and the cage **106**. In this way, electromagnetic waves between the cage **106** and the shell **210** are shielded to prevent the electromagnetic waves from leaking out of the cage **106**.

(65) In addition, since contact between the metal frame **301** of the elastic sleeve **300** and the shell **210** is surface contact, a gap is likely to exist between the metal frame **301** and the shell **210**, which causes the electromagnetic waves to leak out from the gap between the metal frame **301** and the shell **210**. Therefore, the optical module **200** further includes a conductive fiber sheet **400** disposed between the elastic sleeve **300** and the shell **210**. The conductive fiber sheet **400** is configured to connect the shell **210** and the elastic sleeve **300**, so that the elastic sleeve **300** is in sufficient contact with the shell **210** to block the gap between the metal frame **301** and the shell **210** of the optical module **200**. As a result, sufficient contact between the shell **210** of the optical module **200** and the cage **106** of the optical network terminal **100** is further ensured.

(66) The conductive fiber sheet **400** may have elasticity, and the conductive fiber sheet **400** may be deformed by a squeeze. In this way, when the metal frame **301** of the elastic sleeve **300** is clamped into the groove **203**, the metal frame **301** will squeeze the conductive fiber sheet **400**, so that the

conductive fiber sheet **400** blocks the gap between the shell **210** of the optical module **200** and the metal frame **301**. In addition, the conductive fiber sheet **400** has certain elasticity, so that the elastic sleeve **300** may be more easily clamped into the groove **203** of the shell **210** of the optical module **200**.

(67) FIG. **8** is a structural diagram of a conductive fiber sheet in an optical module according to some embodiments. As shown in FIG. **8**, the conductive fiber sheet **400** includes a conductive frame **401** and a conductive elastic member **402**. The conductive frame **401** includes a first conductive sheet **4011**, a second conductive sheet **4012** and a third conductive sheet **4013**. One end of the first conductive sheet **4011** is connected to the second conductive sheet **4012**, and the other end thereof is connected to the third conductive sheet **4013**. The second conductive sheet **4012** and the third conductive sheet **4013** are symmetrically arranged with respect to the first conductive sheet **4011**. That is, the second conductive sheet **4012** and the third conductive sheet **4013** are arranged on a same side of the first conductive sheet **4011**, and an included angle $\beta_{\text{sub.1}}$ between the second conductive sheet **4012** and the first conductive sheet **4011** is equal to an included angle $\beta_{\text{sub.2}}$ between the third conductive sheet **4013** and the first conductive sheet **4011**. For example, each of the second conductive sheet **4012** and the third conductive sheet **4013** is substantially perpendicular to the first conductive sheet **4011**. That is to say, the first conductive sheet **4011**, the second conductive sheet **4012** and the third conductive sheet **4013** constitute a C-shaped conductive member, and the conductive fiber sheet **400** may be sleeved in the groove **203** of the shell **210** of the optical module **200** through an opening at a lower end.

(68) Since the fourth side plate **3014** and the fifth side plate **3015** of the metal frame **301** are provided with an elastic fastener **3016**, it is inconvenient to arrange conductive sheets among the bottom plate **2021** of the lower shell **202**, the fourth side plate **3014** and the fifth side plate **3015**. Therefore, the conductive fiber sheet **400** is composed of three conductive sheets, and an overall structure thereof is a C-shaped structure, so that the conductive fiber sheet **400** may be easily clamped into the groove **203** of the shell **210**.

(69) FIG. **9** is an assembly diagram of a conductive fiber sheet in an optical module and a shell of the optical module according to some embodiments. FIG. **10** is an assembly diagram of an elastic sleeve, a conductive fiber sheet in an optical module and a shell of the optical module according to some embodiments. As shown in FIGS. **9** and **10**, when the conductive fiber sheet **400** is clamped into the groove **203** of the shell **210** of the optical module **200**, the first conductive sheet **4011** of the conductive fiber sheet **400** is clamped into the groove on the cover plate **2011** of the upper shell **201**, the second conductive sheet **4012** is clamped into the grooves on the first upper side plate **2012A** of the upper shell **201** and the first lower side plate **2022A** of the lower shell **202**, and the third conductive sheet **4013** is clamped into the grooves on the second upper side plate **2012B** of the upper shell **201** and the second lower side plate **2022B** of the lower shell **202**.

(70) When the conductive fiber sheet **400** is sleeved in the groove **203** of the shell **210** of the optical module **200**, a thickness of the conductive fiber sheet **400** may be greater than a depth of the groove **203**. That is, the conductive fiber sheet **400** protrudes from a surface of the shell **210** of the optical module **200**. In this way, the conductive fiber sheet **400** can be compressed into the groove **203** by the elastic sleeve **300**, so that a good electrical connection is formed between the elastic sleeve **300** and the shell **210** of the optical module **200**. As a result, the conductive fiber sheet **400** is in sufficient contact with the shell **210** and the elastic sleeve **300**, so as to ensure sufficient connection between the shell **210** of the optical module **200** and the elastic sleeve **300**.

(71) In some embodiments, a dimension of the first conductive sheet **4011** of the conductive fiber sheet **400** may be equal to or slightly less than a dimension of the first side plate **3011** of the metal frame **301**, a dimension of the second conductive sheet **4012** may be equal to or slightly less than a dimension of the second side plate **3012**, and a dimension of the third conductive sheet **4013** may be equal to or slightly less than a dimension of the third side plate **3013**. In this way, a dimension of the conductive fiber sheet **400** is substantially equal to a dimension of the metal frame **301**, so that

the gap between the elastic sleeve **300** and the shell **210** of the optical module **200** may be effectively blocked.

(72) Since contact between the metal frame **301** of the elastic sleeve **300** and the conductive fiber sheet **400** is also surface contact, there may still be a gap between the metal frame **301** and the conductive fiber sheet **400**. Therefore, the metal frame **301** further includes a plurality of concave points **3017**. The plurality of concave points **3017** are uniformly disposed on the first side plate **3011**, the second side plate **3012** and the third side plate **3013**, and the plurality of concave points **3017** are in contact with an outer surface of the conductive fiber sheet **400**. The concave points **3017** may increase reliability of contact between the metal frame **301** and the conductive fiber sheet **400**, and ensure sufficient contact between the metal frame **301** and the conductive fiber sheet **400**.

(73) In some embodiments, the concave points **3017** on the first side plate **3011**, the second side plate **3012** and the third side plate **3013** may be made by stamping process, and the plurality of concave points **3017** are blind holes. As a result, the metal frame **301** and the conductive fiber sheet **400** are fully connected, which makes a resistance between the metal frame **301** and the conductive fiber sheet **400** low.

(74) FIG. **11** is a side view of an assembly of an elastic sleeve, a conductive fiber sheet and a shell of the optical module according to some embodiments. FIG. **12** is an enlarged view of a portion A in FIG. **11**. As shown in FIGS. **11** and **12**, the conductive elastic member **402** is disposed on an end of the first conductive sheet **4011** of the conductive fiber sheet **400**. The conductive elastic member **402** extends substantially in a length direction of the first conductive sheet **4011** (i.e., a direction substantially perpendicular to the second conductive sheet **4012** and the third conductive sheet **4013**), and is located between the metal elastic pieces **302** of the elastic sleeve **300** and the first conductive sheet **4011**. The conductive elastic member **402** and the first conductive sheet **4011** of the conductive frame **401** are arranged in a stepped shape, and the conductive elastic member **402** abuts against the metal elastic pieces **302** for withstanding the metal elastic pieces **302**, so as to ensure that the metal elastic pieces **302** are in contact with the inner surface of the cage **106**.

(75) In some embodiments, the conductive elastic member **402** includes a top surface, a bottom surface, and two arc-shaped connecting surfaces connecting the top surface and the bottom surface. The top surface is in contact with the metal elastic pieces **302**, and the bottom surface is fixed on the first conductive sheet **4011**. In some other embodiments, an end of the first conductive sheet **401** is provided with a conductive boss, and the conductive boss extends in a length direction of the first conductive sheet **4011**. The conductive boss has a through hole inside, an extension direction of the through hole is the same as that of the conductive boss, and the conductive elastic member **402** is inserted into the through hole for withstanding the metal elastic pieces **302**.

(76) In order to withstand the metal elastic pieces **302**, a thickness of the conductive elastic member **402** is not less than a thickness $H_{sub.1}$ of a metal elastic piece **302**. That is, the thickness of the conductive elastic member **402** is greater than or equal to a distance $H_{sub.1}$ from a highest point to a lowest point of the metal elastic piece **302**, so as to withstand the metal elastic piece **302**. When the optical module **200** is inserted into the cage **106**, the inner surface of the cage **106** squeezes the metal elastic pieces **302**. Since the conductive elastic member **402** has elasticity, the metal elastic pieces **302** squeeze the conductive elastic member **402**, and the conductive elastic member **402** squeezes the conductive frame **401**, and the conductive frame **401** squeezes the shell **210** of the optical module **200**. Therefore, a gap between the elastic sleeve **300** and the cage **106**, a gap between the elastic sleeve **300** and the conductive fiber sheet **400**, and a gap between the conductive fiber sheet **400** and the shell **210** can be effectively blocked. As shown in FIG. **12**, after the conductive elastic member **402** is squeezed by the metal elastic pieces **302**, a thickness $H_{sub.2}$ of the conductive elastic member **402** is less than or equal to a thickness $H_{sub.1}$ of a metal elastic piece **302**.

(77) In some embodiments, the conductive elastic member **402** is a conductive foam. That is, a

conductive cloth is wrapped around an outside of the conductive foam. The conductive foam has good elasticity. When the conductive fiber sheet **400** is clamped into the groove **203** of the shell **210**, and then the elastic sleeve **300** is installed, a compression degree of the conductive foam is large, thereby avoiding inconvenience of installing the elastic sleeve **300**.

(78) An assembly process of the optical module provided by some embodiments of the present disclosure is as follows: first, photoelectric devices such as the circuit board **600**, the light emitting device **700**, and the light receiving device **800** of the optical module **200** are installed into the lower shell **202**, and then the upper shell **201** covers the lower shell **202**; then the conductive fiber sheet **400** is clamped into the first groove **2031** on the outer wall of the upper shell **201** and the second groove **2032** on the outer wall of the lower shell **202**; then the elastic sleeve **300** is sleeved on an outside of the conductive fiber sheet **400**, so that the conductive elastic member **402** of the conductive fiber sheet **400** is located between the first conductive sheet **4011** and the metal elastic pieces **302**, and the elastic sleeve **300** is clamped into the clamping grooves **206** of the lower shell **202** through the elastic fasteners **3016** to fix the elastic sleeve **300**.

(79) After the optical module **200** is assembled, when the optical module **200** is inserted into the cage **106** of the optical network terminal **100**, the inner surface of the cage **106** squeezes the metal elastic pieces **302** of the elastic sleeve **300**, and the conductive elastic member **402** under the metal elastic pieces **302** withstands the metal elastic pieces **302**, so that the metal elastic pieces **302** form a good electrical contact with the cage **106**, and the gap between the elastic sleeve **300** and the cage **106** may be eliminated. In addition, the conductive fiber sheet **400** is located between the elastic sleeve **300** and the shell **210** of the optical module **200**, so that the elastic sleeve **300** forms a good contact with the shell **210** of the optical module **200**, and the gap between the elastic sleeve **300** and the shell **210** of the optical module **200** may be eliminated. In this way, a good contact is formed between the shell **210** of the optical module **200** and the cage **106**, thereby forming a metal sealed cavity and achieving efficient shielding. As a result, electromagnetic waves between the shell **210** of the optical module **200** and the cage **106** are shielded, so that electromagnetic waves cannot radiate out of the cage **106** to affect other optoelectronic devices.

(80) FIG. **13** is an assembly diagram of a shell and an unlocking component in an optical module according to some embodiments. FIG. **14** is an exploded structural diagram of a shell and an unlocking component in an optical module according to some embodiments. As shown in FIGS. **13** and **14**, the optical module **200** further includes an unlocking component **500**. The unlocking component **500** is located on an outer wall of the shell **210**, and is configured to implement fixed connection between the optical module **200** and the master monitor, or to release the fixed connection between the optical module **200** and the master monitor.

(81) An end of the unlocking component **500** may be matched and connected to an outer wall of an upper side plate **2012** of the upper shell **201**, or may also be matched and connected to an outer wall of a lower side plate **2022** of the lower shell **202**, or may also be matched and connected to outer walls of the upper side plate **2012** of the upper shell **201** and the lower side plate **2022** of the lower shell **202**. FIG. **14** shows that an end of the unlocking component **500** is matched and connected to outer walls of the upper side plate **2012** of the upper shell **201** and the lower side plate **2022** of the lower shell **202**.

(82) As shown in FIG. **14**, the unlocking component **500** includes a handle **510**, a first unlocking cantilever **520** and a second unlocking cantilever **530**. An end of the handle **510** is connected to one end of the first unlocking cantilever **520** and one end of the second unlocking cantilever **530**, the other end of the first unlocking cantilever **520** is matched and connected to outer walls of the first upper side plate **2012A** of the upper shell **201** and the first lower side plate **2022A** of the lower shell **202**, and the other end of the second unlocking cantilever **530** is matched and connected to outer walls of the second upper side plate **2012B** of the upper shell **201** and the second lower side plate **2022B** of the lower shell **202**. The handle **510** is dragged, thereby driving the first unlocking cantilever **520** and the second unlocking cantilever **530** to move in an extension direction of the

upper side plate **2012** of the upper shell **201** and the lower side plate **2022** of the lower shell **201**. An arrangement of the handle **510** facilitates usage of the unlocking component **500**. In some embodiments of the present disclosure, the first unlocking cantilever **520** and the second unlocking cantilever **530** are a metal member, such as a metal sheet component.

(83) When materials of the outer wall of the shell **210** and the unlocking component **500** are relatively hard materials such as metal materials, and the hard materials are in direct contact with each other, and the contact is hard contact. When the outer wall of the shell **210** is in hard contact with the unlocking component **500**, a gap will exist between the shell **210** and the unlocking component **500**, so that electromagnetic waves radiated from the electrical port **204** of the optical module **200** may propagate along the gap. In order to prevent the electromagnetic waves from radiating and transmitting along the gap between the outer wall of the shell **210** and the unlocking component **500**, as shown in FIGS. **15** and **19**, the optical module **200** further includes at least one conductive fiber sheet **610**. The at least one conductive fiber sheet **610** is disposed on the outer wall of the shell **210**, the unlocking component **500** covers the at least one conductive fiber sheet **610**, the shell **210** and the unlocking component **500** are connected by the at least one conductive fiber sheet **610**, and the at least one conductive fiber sheet **610** is configured to seal the gap between the shell **210** and the unlocking component **500**. Some embodiments of the present disclosure do not limit the number of the at least one conductive fiber sheet **610**, which may be one, two or more.

(84) In some embodiments of the present disclosure, as shown in FIGS. **15** and **19**, the conductive fiber sheet **610** is a conductive pad **611** disposed on the outer wall of the shell **210**, or is a convex rib **612** formed on the outer wall of the shell **210**.

(85) The conductive fiber sheet **610** may be disposed on the outer wall of the upper side plate **2012** of the upper shell **201**, or may be disposed on the outer wall of the lower side plate **2022** of the lower shell **202**, or may further be disposed on the outer walls of the upper side plate **2012** of the upper shell **201** and the lower side plate **2022** of the lower shell **202**. FIGS. **15** and **19** show that the conductive fiber sheet **610** is disposed on outer walls of the upper side plate **2012** of the upper shell **201** and the lower side plate **2022** of the lower shell **202**.

(86) In the optical module provided by some embodiments of the present disclosure, the outer wall of the shell **210** is provided with the conductive fiber sheet **610**. On the one hand, the shell **210** and the unlocking component **500** are connected by the conductive fiber sheet **610**. On the other hand, formation of a gap between the shell **210** and the unlocking component **500** is avoided, thereby preventing the electromagnetic waves from radiating and propagating from the gap between the shell **210** and the unlocking component **500**. Thus, electromagnetic shielding effect of the optical module **200** is improved.

(87) As shown in FIG. **16**, in order to facilitate installation of the unlocking component **500** and ensure flatness of an outer contour of the optical module **200**, the shell **210** has at least one recessed area **211**. The at least one recessed area **211** is disposed on outer walls of the upper side plate **2012** and the lower side plate **2022**, and the recessed area **211** is configured to be matched and connected to an end of the unlocking component **500**. For example, the recessed area **211** is matched and connected to an end of the first unlocking cantilever **520** away from the handle **510** or an end of the second unlocking cantilever **530** away from the handle **510**. At least one conductive fiber sheet **610** is disposed in a corresponding recessed area **211**. Some embodiments of the present disclosure do not limit the number of the at least one recessed area **211**, which may be one, two or more.

(88) FIG. **15** is an assembly diagram of a shell and an unlocking component in an optical module according to some embodiments. FIG. **16** is an assembly diagram of a conductive pad and a shell in an optical module according to some embodiments. As shown in FIGS. **15** and **16**, the conductive fiber sheet **610** is a conductive pad **611**, and at least one conductive pad **611** is disposed on outer walls of the upper side plate **2012** of the upper shell **201** and the lower side plate **2022** of the lower shell **202**. The unlocking component **500** covers the at least one conductive pad **611**. For example, the first unlocking cantilever **520** or the second unlocking cantilever **530** covers the at least one

conductive pad **611**, and the at least one conductive pad **611** is configured to connect the first unlocking cantilever **520**, the upper side plate **2012** of the upper shell **201** and the lower side plate **2022** of the lower shell **202**, or to connect the second unlocking cantilever **530**, the upper side plate **2012** of the upper shell **201** and the lower side plate **2022** of the lower shell **202**.

(89) In some embodiments of the present disclosure, a thickness of the conductive pad **611** is in a range of 0.1 mm to 0.3 mm, for example, 0.1 mm, 0.15 mm, 0.2 mm, 0.25 mm, or 0.3 mm. The conductive pad **611** may be a flexible conductive cloth, a flexible wave-absorbing material pad, or the like, and the conductive pad **611** may be attached to the outer wall of the shell **210**.

(90) As mentioned above, if the outer wall of the shell **210** made of metal materials is in direct contact with the unlocking component **500** made of metal materials, contact between the outer wall of the shell **210** and the unlocking component **500** is hard contact, and a gap usually exists between objects with hard contact. When the optical module **200** is inserted into the cage of the master monitor, electromagnetic waves radiated from the electrical port **204** of the optical module **200** may radiate and propagate along the gap between the outer wall of the shell **210** and the unlocking component **500**. In the optical module provided by some embodiments of the present disclosure, the conductive pad **611** is disposed between the shell **210** and the unlocking component **500**, and the shell **210** and the unlocking component **500** are connected by the conductive pad **611**, so as to avoid forming a gap between the shell **210** and the unlocking component **500**. Thus, electromagnetic waves are prevented from radiating and propagating from the gap between the shell **210** and the unlocking component **500**, and the electromagnetic shielding effect of the optical module **200** is improved.

(91) In some embodiments of the present disclosure, the at least one conductive pad **611** includes two conductive pads **611**, which are a first conductive pad **611A** and a second conductive pad **611B**. As shown in FIGS. **16** to **18**, the first conductive pad **611A** is disposed on the outer walls of the first upper side plate **2012A** and the first lower side plate **2022A**. The first unlocking cantilever **520** covers the first conductive pad **611A**, and the first conductive pad **611A** is fixed between the first unlocking cantilever **520** and the outer walls of the first upper side plate **2012A** and the first lower side plate **2022A**. The first conductive pad **611A** connects the first unlocking cantilever **520** and the outer walls of the first upper side plate **2012A** and the first lower side plate **2022A**, so that the first conductive pad **611A** seals the gap between the first unlocking cantilever **520** and the outer walls of the first upper side plate **2012A** and the first lower side plate **2022A**, thereby preventing electromagnetic waves from radiating out from the gap between the first unlocking cantilever **520** and the outer walls of the first upper side plate **2012A** and the first lower side plate **2022A**.

(92) FIG. **17** is an exploded diagram of a shell, a conductive pad, and an unlocking component in an optical module according to some embodiments. As shown in FIG. **17**, the second conductive pad **611B** is disposed on the outer walls of the second upper side plate **2012B** and the second lower side plate **2022B**. The second unlocking cantilever **530** covers the second conductive pad **611B**, and the second conductive pad **611B** is fixed between the second unlocking cantilever **530** and the outer walls of the second upper side plate **2012B** and the second lower side plate **2022B**. The second conductive pad **611B** connects the second unlocking cantilever **530** and the outer walls of the second upper side plate **2012B** and the second lower side plate **2022B**, so that the second conductive pad **611B** seals the gap between the second unlocking cantilever **530** and the outer walls of the second upper side plate **2012B** and the second lower side plate **2022B**, thereby preventing electromagnetic waves from radiating out from the gap between the second unlocking cantilever **530** and the outer walls of the second upper side plate **2012B** and the second lower side plate **2022B**.

(93) For example, as shown in FIGS. **16** and **17**, the shell **210** has two recessed areas **211**, which are a first recessed area **211A** and a second recessed area **211B**. The first recessed area **211A** is disposed on the first upper side plate **2012A** and the first lower side plate **2022A**, the first recessed area **211A** is configured to be matched and connected to the first unlocking cantilever **520**, and the

first conductive pad **611A** is disposed in the first recessed area **211A**. The second recessed area **211B** is disposed on the second upper side plate **2012B** and the second lower side plate **2022B**, the second recessed area **211B** is configured to be matched and connected to the second unlocking cantilever **530**, and the second conductive pad **611B** is disposed in the second recessed area **211B**. (94) The first conductive pad **611A** seals the gap among the first unlocking cantilever **520**, the first upper side plate **2012A** and the first lower side plate **2022A**, and then the first conductive pad **611A** is connected to the first upper side plate **2012A**, the first lower side plate **2022A** and the first unlocking cantilever **520**. The second conductive pad **611B** seals the gap among the second unlocking cantilever **530**, the second upper side plate **2012B** and the second lower side plate **2022B**, and the second conductive pad **611B** is connected to the second upper side plate **2012B**, the second lower side plate **2022B** and the second unlocking cantilever **530**. In this way, by arranging the first conductive pad **611A** among the first upper side plate **2012A**, the first lower side plate **2022A** and the first unlocking cantilever **520**, and arranging the second conductive pad **611B** among the second upper side plate **2012B**, the second lower side plate **2022B** and the second unlocking cantilever **530**, it is possible to ensure that the first unlocking cantilever **520** is in sufficient contact with the outer walls of the first upper side plate **2012A** and the first lower side plate **2022A**, and the second unlocking cantilever **530** is in sufficient contact with the outer walls of the second upper side plate **2012B** and the second lower side plate **2022B**, to prevent electromagnetic waves from radiating and transmitting from the gap among the first upper side plate **2012A**, the first lower side plate **2022A** and the first unlocking cantilever **520** and the gap among the second upper side plate **2012B**, the second lower side plate **2022B** and the second unlocking cantilever **530**, and to fully improve the electromagnetic shielding effect of the optical module **200**.

(95) In some embodiments of the present disclosure, a middle portion of the first unlocking cantilever **520** covers the first conductive pad **611A**, and a middle portion of the second unlocking cantilever **530** covers the second conductive pad **611B**. In some embodiments of the present disclosure, the middle portion of the first unlocking cantilever **520** squeezes and is in contact with the first conductive pad **611A**, and the middle portion of the second unlocking cantilever **530** squeezes and is in contact with the second conductive pad **611B**.

(96) In some embodiments of the present disclosure, a width of the first conductive pad **611A** is greater than or equal to a width of the middle portion of the first unlocking cantilever **520**, and a width of the second conductive pad **611B** is greater than or equal to a width of the middle portion of the second unlocking cantilever **530**. In some embodiments of the present disclosure, the width of the first conductive pad **611A** is equal to the width of the middle portion of the first unlocking cantilever **520**, and the width of the second conductive pad **611B** is equal to the width of the middle portion of the second unlocking cantilever **530**.

(97) In order to facilitate connection between the first unlocking cantilever **520** and the first upper side plate **2012A** and unlocking of the optical module **200** and the cage **106**, the unlocking component **500** further includes a first locking hook **521** disposed on an end of the first unlocking cantilever **520** away from the handle **510**, and the shell **210** further has a first locking slot **212A** disposed on the outer wall of the first upper side plate **2012A**. When the unlocking component **500** is not dragged, the first locking hook **521** is embedded in the first locking slot **212A**. When the unlocking component **500** is dragged, the first locking hook **521** moves in the first locking slot **212A** along a dragging direction. When the unlocking component **500** is released after being dragged, the first locking hook **521** returns to an initial position in the first locking slot **212A**. Therefore, the first locking slot **212A** may be used to accommodate the first locking hook **521** and limit movement of the first locking hook **521**.

(98) In some embodiments of the present disclosure, the unlocking component **500** further includes a first protrusion **522** and a third protrusion **523** disposed on an end of the first unlocking cantilever **520** away from the handle **510**. The first protrusion **522** is located on one side of the first unlocking

cantilever **520**, and the third protrusion **523** is located on the other side of the first unlocking cantilever **520**. In some embodiments of the present disclosure, the shell **210** further has a first position-limiting groove **213** disposed on the first upper side plate **2012A** and a third position-limiting groove **214** disposed on the first lower side plate **2022A**. The first protrusion **522** is embedded in the first position-limiting groove **213**, and the third protrusion **523** is embedded in the third position-limiting groove **214**. When the unlocking component **500** is dragged or the unlocking component **500** is released after being dragged, the first protrusion **522** moves in the first position-limiting groove **213**, and the third protrusion **523** moves in the third position-limiting groove **214**. Cooperation of the first protrusion **522** with the first position-limiting groove **213** and cooperation of the third protrusion **523** with the third position-limiting groove **214** make it more convenient to achieve a positional limit on unlocking movement of the first unlocking cantilever **520** and ensure stability of usage of the unlocking component **500**.

(99) In some embodiments of the present disclosure, in order to facilitate connection between the second unlocking cantilever **530** and the second upper side plate **2012B** and unlocking of the optical module **200** and the cage **106**, the unlocking component **500** further includes a second locking hook **531** disposed on an end of the second unlocking cantilever **530** away from the handle **510**, and the shell **210** further has a second locking slot **212B** disposed on the outer wall of the second upper side plate **2012B**. When the unlocking component **500** is not dragged, the second locking hook **531** is embedded in the second locking slot **212B**. When the unlocking component **500** is dragged, the second locking hook **531** moves in the second locking slot **212B** along a dragging direction. When the unlocking component **500** is released after being dragged, the second locking hook **531** returns to an initial position in the second locking slot **212B**. Therefore, the second locking slot **212B** may be used to accommodate the second locking hook **531** and limit movement of the second locking hook **531**.

(100) In some embodiments of the present disclosure, during unlocking of the optical module **200**, the handle **510** is dragged, and the first locking hook **521** and the second locking hook **531** are used to lift up a locking member on the cage of the master monitor, so that the optical module **200** may be unlocked and detached from the cage of the master monitor.

(101) In some embodiments of the present disclosure, the unlocking component **500** further includes a second protrusion **532** and a fourth protrusion **533** disposed on an end of the second unlocking cantilever **530** away from the handle **510**, the second protrusion **532** is located on one side of the second unlocking cantilever **530**, and the fourth protrusion **533** is located on the other side of the second unlocking cantilever **530**. In some embodiments of the present disclosure, the shell **210** further has a second position-limiting groove **215** disposed on the second upper side plate **2012B** and a fourth position-limiting groove **216** disposed on the second lower side plate **2022B**. The second protrusion **532** is embedded in the second position-limiting groove **215**, and the fourth protrusion **533** is embedded in the fourth position-limiting groove **216**. When the unlocking component **500** is dragged or the unlocking component **500** is released after being dragged, the second protrusion **532** moves in the second position-limiting groove **215**, and the fourth protrusion **533** moves in the fourth position-limiting groove **216**. Cooperation of the second protrusion **532** with the second position-limiting groove **215** and cooperation of the fourth protrusion **533** with the fourth position-limiting groove **216** make it more convenient to achieve a positional limit on unlocking movement of the second unlocking cantilever **530** and ensure the stability of usage of the unlocking component **500**.

(102) In some embodiments of the present disclosure, the outer walls of the first upper side plate **2012A** and the first lower side plate **2022A** have a first mounting surface **217A**, the first mounting surface **217A** is a bottom surface of the first recessed area **211A**, and the first conductive pad **611A** is attached to the first mounting surface **217A**. An arrangement position of the first mounting surface **217A** should avoid interfering with the unlocking component **500**. An arrangement of the first mounting surface **217A** facilitates installation and arrangement of the first conductive pad

611A, ensures installation accuracy of the first conductive pad **611A**, and avoids improper installation of the first conductive pad **611A**, which affects usage of the unlocking component **500**. For example, in order to prevent improper installation of the first conductive pad **611A** from affecting movement of the first unlocking cantilever **520** relative to the shell **210**, the first mounting surface **217A** is away from the first locking slot **212A**. In some embodiments of the present disclosure, a position of the first mounting surface **217A** may correspond to that of a clamping member of the cage of the master monitor clamping the optical module, and the clamping member of the cage of the master monitor clamping the optical module may squeeze the first unlocking cantilever **520** when clamping the optical module, so that the first unlocking cantilever **520** is in sufficient contact with the first conductive pad **611A**.

(103) In some embodiments of the present disclosure, the outer walls of the second upper side plate **2012B** and the second lower side plate **2022B** have a second mounting surface **217B**, the second mounting surface **217B** is a bottom surface of the second recessed area **211B**, and the second conductive pad **611B** is attached to the second mounting surface **217B**. An arrangement position of the second mounting surface **217B** should avoid interfering with the unlocking component **500**. An arrangement of the second mounting surface **217B** facilitates installation and arrangement of the second conductive pad **611B**, ensures installation accuracy of the second conductive pad **611B**, and avoids improper installation of the second conductive pad **611B**, which affects usage of the unlocking component **500**. For example, in order to prevent improper installation of the second conductive pad **611B** from affecting movement of the second unlocking cantilever **530** relative to the shell **210**, the second mounting surface **217B** is away from the second locking slot **212B**. In some embodiments of the present disclosure, a position of the second mounting surface **217B** may correspond to that of a clamping member of the cage of the master monitor clamping the optical module, and then the clamping member of the cage of the master monitor clamping the optical module may squeeze the second unlocking cantilever **530** when clamping the optical module **200**, so that the second unlocking cantilever **530** is in sufficient contact with the second conductive pad **611B**.

(104) In some embodiments of the present disclosure, the first mounting surface **217A** is disposed on the first lower side plate **2022A**, and the second mounting surface **217B** is disposed on the second lower side plate **2022B**; in some other embodiments of the present disclosure, the first mounting surface **217A** is disposed on the first upper side plate **2012A**, and the second mounting surface **217B** is disposed on the second upper side plate **2012B**.

(105) In some embodiments of the present disclosure, as shown in FIGS. **16** and **17**, the shell **210** has a first spring groove **218A** and a second spring groove **218B**, and the first spring groove **218A** is disposed on an end of the first upper side plate **2012A** of the upper shell **201** close to the optical port **205**, and the second spring groove **218B** is disposed on an end of the second upper side plate **2012B** of the upper shell **201** close to the optical port **205**. The optical module **200** further includes a first spring **219A** and a second spring **219B**, the first spring **219A** is disposed in the first spring groove **218A**, and the second spring **219B** is disposed in the second spring groove **218B**. The unlocking component **500** further includes a first spring hook **524** and a second spring hook **534**, the first spring hook **524** is disposed on an end of the first unlocking cantilever **520** connected to the handle **510**, and the second spring hook **534** is disposed on an end of the second unlocking cantilever **530** connected to the handle **510**.

(106) The first spring hook **524** is matched and connected to the first spring **219A**, and the second spring hook **534** is matched and connected to the second spring **219B**. Cooperation between the first spring hook **524** and the first spring **219A** and cooperation between the second spring hook **534** and the second spring **219B** may achieve automatic recovery after the unlocking component **500** unlocks the optical module **200**.

(107) FIG. **19** is another structural diagram of a shell in an optical module, in accordance with some embodiments. As shown in FIG. **19**, the conductive fiber sheet **610** is a convex rib **612**. At

least one conductive fiber sheet **610** includes a plurality of convex ribs **612**, and the plurality of convex ribs **612** are disposed on the outer wall of the shell **210**.

(108) In some embodiments of the present disclosure, a plurality of first convex ribs **612A** are disposed on the outer wall of the first upper side plate **2012A**. The plurality of first convex ribs **612A** are connected to the first unlocking cantilever **520** and the first upper side plate **2012A**, and are configured to seal the gap between the first unlocking cantilever **520** and the first upper side plate **2012A**. A plurality of second convex ribs **612B** are disposed on the first lower side plate **2022A**. The plurality of second convex ribs **612B** are connected to the first unlocking cantilever **520** and the first lower side plate **2022A**, and are configured to seal the gap between the first unlocking cantilever **520** and the first lower side plate **2022A**.

(109) When the optical module **200** is inserted into the master monitor for use, electromagnetic radiation in the master monitor is transmitted to the plurality of first convex ribs **612A** and the plurality of second convex ribs **612B**, and the plurality of first convex ribs **612A** and the plurality of second convex ribs **612B** seal the gap between the first unlocking cantilever **520** and the shell **210**. Thus, electromagnetic waves are prevented from radiating out from the gap between the first unlocking cantilever **520** and the shell **210**. In addition, grooves formed between adjacent convex ribs may reflect and absorb the electromagnetic radiation for multiple times to further improve the electromagnetic shielding effect. Therefore, effect of improving electromagnetic shielding of the master monitor may be achieved through the plurality of first convex ribs **612A** and the plurality of second convex ribs **612B**.

(110) In some embodiments of the present disclosure, a plurality of third convex ribs may be disposed on the second upper side plate **2012B**, and a plurality of fourth convex ribs may be disposed on the second lower side plate **2022B**. For structures and functions of the plurality of third convex ribs and the plurality of fourth convex ribs, reference may be made to those of the plurality of first convex ribs **612A** and the plurality of second convex ribs **612B**.

(111) In some embodiments of the present disclosure, a height of each rib on the outer walls of the first upper side plate **2012A**, the first lower side plate **2012B**, the second upper side plate **2022A** and the second lower side plate **2022B** is in a range of 0.05 to 0.15 mm, a width thereof is in a range of 0.5 to 1.5 mm, and spacing between adjacent convex ribs is in a range of 0.5 to 1.5 mm. In this way, it is convenient to further ensure the electromagnetic shielding effect.

(112) The foregoing descriptions are merely specific implementations of the present disclosure, but the protection scope of the present disclosure is not limited thereto. Changes or replacements that any person skilled in the art could conceive of within the technical scope of the present disclosure shall be included in the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure shall be subject to the protection scope of the claims.

Claims

1. An optical module, comprising: a shell having a groove disposed on an outer wall thereof; an elastic sleeve including a metal frame and a plurality of metal elastic pieces, the metal frame being sleeved in the groove and fixedly connected to the shell; the plurality of metal elastic pieces being disposed on an edge of the metal frame and extending to an outside of the metal frame; and a conductive fiber sheet disposed between the shell and the elastic sleeve to connect the shell and the elastic sleeve, wherein the conductive fiber sheet includes: a conductive frame located between the shell and the metal frame; and a conductive elastic member disposed on an edge of the conductive frame and arranged in a stepped shape with the conductive frame, and the conductive elastic member abutting against the plurality of metal elastic pieces.
2. The optical module according to claim 1, wherein a thickness of the conductive frame is greater than a depth of the groove.
3. The optical module according to claim 1, wherein a thickness of the conductive elastic member

is greater than or equal to a thickness of a metal elastic piece.

4. The optical module according to claim 1, wherein the metal frame includes a first side plate, a second side plate, a third side plate, a fourth side plate and a fifth side plate; an end of the first side plate is connected to the second side plate, and another end of the first side plate is connected to the third side plate; an end of the second side plate away from the first side plate is connected to the fourth side plate, and an end of the third side plate away from the first side plate is connected to the fifth side plate; and a gap exists between the fourth side plate and the fifth side plate, and the metal frame is sleeved on an outside of the conductive frame through the gap.

5. The optical module according to claim 4, wherein the plurality of metal elastic pieces are disposed on edges of the first side plate, the second side plate, the third side plate, the fourth side plate and the fifth side plate.

6. The optical module according to claim 5, wherein the plurality of metal elastic pieces extend to an outside of the elastic sleeve along the edges of the first side plate, the second side plate, the third side plate, the fourth side plate and the fifth side plate, and are arranged in an umbrella shape.

7. The optical module according to claim 4, wherein the metal frame includes a first elastic fastener and a second elastic fastener, the first elastic fastener is disposed on the fourth side plate, and the second elastic fastener is disposed on the fifth side plate; and the shell has a first clamping groove and a second clamping groove, and the first elastic fastener and the second elastic fastener are fixedly connected to the first clamping groove and the second clamping groove respectively.

8. The optical module according to claim 4, wherein the metal frame further includes a plurality of concave points; and the plurality of concave points are disposed on the first side plate, the second side plate and the third side plate, and are in contact with an outer surface of the conductive fiber sheet.

9. The optical module according to claim 8, wherein the plurality of concave points are blind holes.

10. The optical module according to claim 4, wherein the conductive frame includes a first conductive sheet, a second conductive sheet and a third conductive sheet, an end of the first conductive sheet is connected to the second conductive sheet, and another end of the first conductive sheet is connected to the third conductive sheet; and the first conductive sheet is located between the shell and the first side plate, the second conductive sheet is located between the shell and the second side plate, and the third conductive sheet is located between the shell and the third side plate.

11. The optical module according to claim 10, wherein the conductive elastic member is disposed on the first conductive sheet and extends in a length direction of the first conductive sheet.

12. The optical module according to claim 10, wherein the second conductive sheet and the third conductive sheet are arranged on a same side of the first conductive sheet, and an included angle between the second conductive sheet and the first conductive sheet is equal to an included angle between the third conductive sheet and the first conductive sheet.

13. The optical module according to claim 10, wherein a dimension of the first conductive sheet is equal to or slightly less than a dimension of the first side plate, a dimension of the second conductive sheet is equal to or slightly less than a dimension of the second side plate, and a dimension of the third conductive sheet is equal to or slightly less than a dimension of the third side plate.

14. The optical module according to claim 4, wherein the second side plate and the third side plate are disposed on a same side of the first side plate, and an included angle between the second side plate and the first side plate is equal to an included angle between the third side plate and the first side plate; and the fourth side plate and the fifth side plate are located in a same plane, the fourth side plate and the fifth side plate both extend toward an inside of the elastic sleeve, and an included angle between the fourth side plate and the second side plate is equal to an included angle between the fifth side plate and the third side plate.

15. The optical module according to claim 1, wherein the groove is configured to surrounds the

outer wall of the shell; the shell includes a lower shell and an upper shell covering the lower shell; the lower shell includes a bottom plate, and two lower side plates located on two sides of the bottom plate and disposed perpendicular to the bottom plate; and the upper shell includes a cover plate, and two upper side plates located on two sides of the cover plate and disposed perpendicular to the cover plate; and the metal frame is fixedly connected to the bottom plate of the lower shell.

16. The optical module according to claim 15, wherein the groove includes a first groove and a second groove; the first groove is disposed on outer walls of the cover plate and the two upper side plates; and the second groove is disposed on outer walls of the bottom plate and the two lower side plates.
